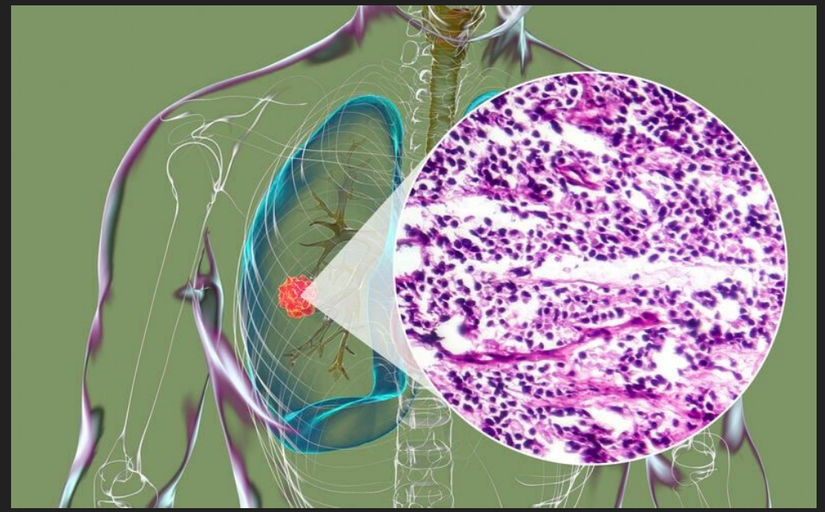
EGFR Duplication Variants



Group C

Mir Syed, Emily Moya, Simran Malik

Introduction

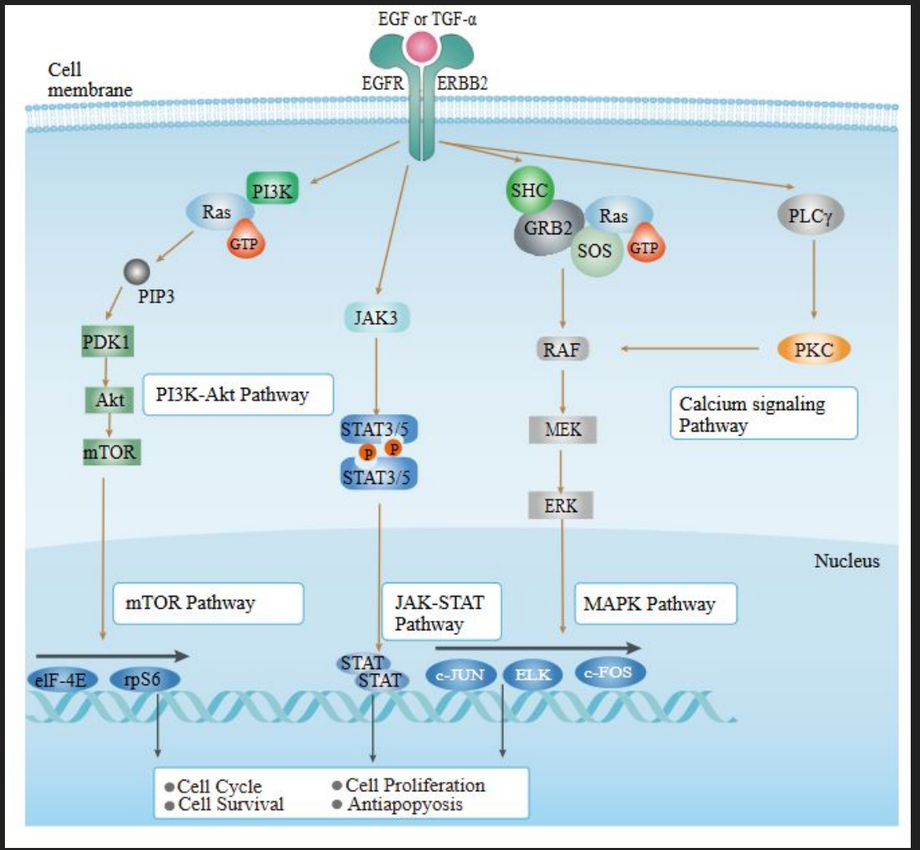
Given Observation:

- "You are working with primary lung cancer cells derived from patient tumors in the laboratory to test new drug therapies and you use sequencing to identify genomics changes in these cells. In one patient, you observe a small duplication of six nucleotides in the growth factor gene *EGFR*. In another patient, you notice there are six additional copies of the entire gene *EGFR*. What is the difference between these types of duplications?"

Red Text = Key Concepts

Hallmarks of Cancer

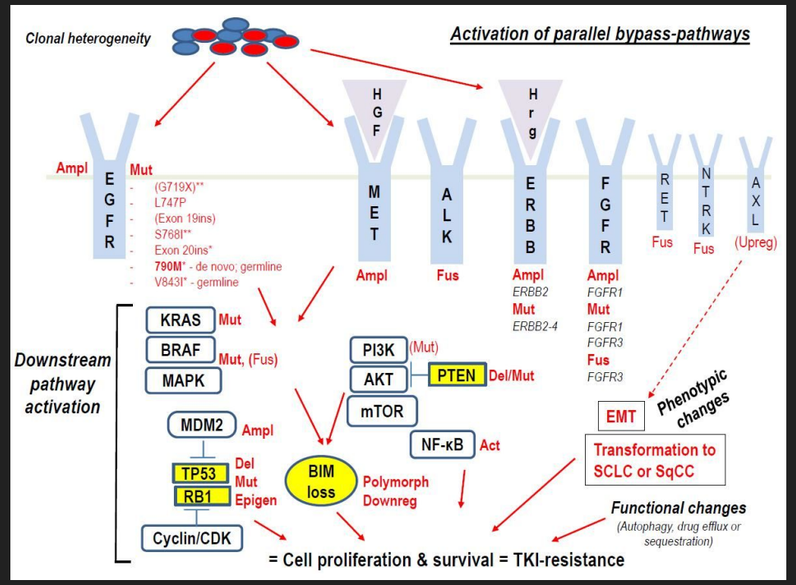
- EGFR is a gene that provides instructions to make an epidermal growth factor receptor. It is a cell surface protein that binds to epidermal growth factors. These epidermal growth factors play and important role in regulating cell growth, division, and survival. In normal function, it can stimulate epidermal and other cells to proliferate through the PI3K/Akt, MAPK/ERK, and JAK/STAT pathways (Albert et. al, 2023).
- EGFR is related to the hallmark of "sustained proliferative signaling". When EGFRs function is altered, it can lead to uncontrolled cell growth and survival independent of external signals.
- EGFR mutations also promote metastasis through repeat paracrine loops. This repeat loops control critical steps in the metastasis pathway. (Uribe et. al, 2021).
- The small duplication of six nucleotides may have caused a frameshift mutation in the EGFR. Depending on the location, this could alter the activation of EGFRs to be permanently on (Improta et. al, 2018).
- The gene amplification of the entire EGFR gene could lead to the overexpression of the EGFR protein on the cell surface. This can enhance the cell's sensitivity to growth factors.



Graphic Source: Creative BioLabs, Non-small Cell Lung Cancer, 2023

How EGFR Changes Lead to Cancer Growth

Mechanism/Category	6 Nucleotide Small Duplication (Patient A)	Large scale gene amplification (Patient B)
What it is	A tiny repeat inside a gene (Just 6 DNA letters duplicated such as A,T,C,G)	The whole EGFR gene is copied 6 times rather than only once
Effect on the Protein (shape or function changes)	Changes how the EGFR protein is shaped	Since there's too many copies of the protein, the cell is flooded with growth signals
What happens	The protein is always “on” so it tells the cell to grow even when it should not	The cell becomes super sensitive to growth signals (due to an enhanced ligand)
How it Causes Cancer	Always going or “on”	giving the cell too many signals which means more chances to “go” causing overexpression
Treatment impact	May work with some cancer drugs	May need a higher dosage of drugs, compared to small duplication, such as antibodies



Mechanisms of EGFR-driven TKI resistance and bypass pathway activation in cancer. From MDPI.com

This diagram shows how mutations or amplifications of the EGFR gene causes cancer growth and resistance against EGFR inhibitors (which are called TKI’s), in cases when EGFR is blocked by treatment and drugs cancer cells often find backup pathways, or completely change their behavior, so they can continue to grow. This shows why identifying specific EGFR changes is super important in predicting treatment response to personalize treatment and therapies.

References

- Alberts, B., Heald, R., Hopkin, K., Johnson, A., Morgan, D., Roberts, K., & Walter, P. (2023). *Essential Cell Biology* (6th ed.). W.W. Norton & Company.
- Improta, G., Zupa, A., Natalicchio, M. I., Sisinni, L., Marinaccio, A., Bozza, G., Vita, G., Aieta, M., & Landriscina, M. (2018). Uncommon frame-shift exon 19 EGFR mutations are sensitive to EGFR tyrosine kinase inhibitors in non-small cell lung carcinoma. Medical oncology (Northwood, London, England), 35(3), 28. https://doi.org/10.1007/s12032-018-1078-7
- Moffitt Cancer Center. (n.d.). *Small cell lung cancer treatment | moffitt*. Moffitt Cancer Center. https://www.moffitt.org/cancers/lung-cancer/faqs/what-are-treatment-options-for-small-cell-lung-cancer/
- Non-small cell lung cancer overview. (2023, August 26). Creative Biolabs. https://www.creativebiolabs.net/non-small-cell-lung-cancer-overview-signaling-pathway-diagnosis-targeted-therapy.htm
- Santoni-Rugiu, Melchior, Urbanska, Jakobsen, Stricker, Grauslund, & Sørensen. (2019). Intrinsic resistance to EGFR-Tyrosine Kinase Inhibitors in EGFR-Mutant Non-Small Cell Lung Cancer: Differences and Similarities with Acquired Resistance. *Cancers*, *11*(7), 923. https://doi.org/10.3390/cancers11070923
- Uribe, Mary Luz, et al. “EGFR in Cancer: Signaling Mechanisms, Drugs, and Acquired Resistance.” *Cancers*, vol. 13, no. 11, 1 June 2021, p. 2748, pmc.ncbi.nlm.nih.gov/articles/PMC8197917/, https://doi.org/10.3390/cancers13112748.